

Image Processing Project: Feature-Based Image Processing and Reinforcement Learning for CarRacing-v3

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Abstract—This project studies an image-processing-centric pipeline for Gymnasium CarRacing-v3, a continuous-control driving task where the agent observes a 96×96 RGB frame and emits steering, gas, and brake commands. Rather than training a policy directly on pixels, a deterministic wrapper converts each frame into an HSV-based road mask, cleans it morphologically, and casts nine road-distance rays across a calibrated angular fan. The resulting compact observation is a 16-dimensional vector (nine ray distances, three smoothed previous actions, speed, a curvature proxy, a front-ray change term, and a left-right asymmetry term); four consecutive observations are temporally stacked into a 64-dimensional vector. The same feature pipeline is shared by two downstream evaluators—a completed Stable-Baselines3 PPO baseline trained for 500,000 steps and a SAC fast-result branch validated at a 400,000-step best checkpoint. The PPO baseline reaches a validated raw reward of 938.87 ± 7.86 at 500,000 steps (best parsed 939.53 ± 4.09 at 480,000), and the SAC branch reaches 938.51 ± 4.88 at 400,000. The contribution is the visual representation; the reinforcement-learning algorithms are evaluation tools. The method is sample-efficient and promising, but it is not uniformly robust, and CarRacing-v3 is not claimed to be solved.

Index Terms—image processing, HSV road mask, ray features, CarRacing-v3, PPO, SAC, feature-based reinforcement learning

I. INTRODUCTION

Gymnasium CarRacing-v3 is a compact but demanding continuous-control benchmark [1]. The observation is a 96×96 RGB image, and the action is a three-dimensional continuous vector for steering, gas, and brake. A policy trained directly on pixels must learn visual perception, local road geometry, throttle control, and recovery behavior simultaneously. That full end-to-end problem is attractive as a benchmark, but it is expensive for a course project where the training budget, hardware time, and debugging time are limited.

This project takes a narrower and more reproducible path. The core idea is to keep the reinforcement-learning algorithm standard and move part of the perception burden into a deterministic image-processing wrapper. Each RGB observation is cropped, converted to HSV, thresholded into a road mask, cleaned morphologically, and summarized by nine distance rays. The policy does not receive the full image; it receives a compact vector describing local road openness, recent controls, speed, and curvature. The resulting observation is small enough to train with an MLP policy rather than a convolutional image policy.

The same compact visual representation is then used as a fixed downstream input to two standard off-the-shelf learners

from Stable-Baselines3 [2], [3], [4]: a completed PPO [5] baseline and a SAC fast-result branch [6]. Holding the perception layer constant and varying only the policy optimizer makes the image-processing contribution the isolated variable of the study.

The design is intentionally conservative. It does not introduce a custom reinforcement-learning variant, a new neural-network architecture, or a learned perception module. The wrapper is deterministic image processing, the learners are unmodified Stable-Baselines3 algorithms, and the final evaluation reloads saved artifacts from disk. This makes the project auditable: mask thresholds, ray geometry, action smoothing, reward shaping, algorithm settings, normalization statistics, and evaluation seeds are all explicit.

The main empirical results are strong but must be read carefully. The PPO baseline is a fully completed 500,000-step run reaching 938.87 ± 7.86 raw reward (best parsed 939.53 ± 4.09). The SAC branch is a *partial* fast-result run whose validated number is the 400,000-step best checkpoint (938.51 ± 4.88); it did not complete 500,000 steps. The PPO/SAC comparison is therefore not compute-equivalent, and SAC is not claimed to beat PPO. The results support the claim that the compact road-geometry representation is sample-efficient and promising; they do not support a claim that the environment is solved or that the method is uniformly robust.

The remainder of the paper defines the environment and problem, describes the image-processing pipeline and the compact feature representation in detail, presents the evaluation setup, reports the PPO baseline and SAC fast-result experiments, analyzes the results, and discusses limitations.

II. BACKGROUND AND RELATED WORK

Two broad strategies exist for visual control in CarRacing-v3. The first is end-to-end pixel control, in which a convolutional policy is trained directly on the rendered frame and must jointly learn perception and control; this is the setting most deep reinforcement-learning baselines target [5], [6]. It is general but sample-hungry and hard to debug, because perception errors and control errors are entangled inside a single network. The second strategy, taken here, front-loads perception with deterministic image processing and leaves only control to the learner. Classical color-space thresholding and morphological cleanup are long-standing tools for road

and free-space segmentation [7], and casting distance rays over a free-space mask is a compact way to encode local geometry without a learned encoder. Off-the-shelf implementations of PPO and SAC from Stable-Baselines3 [2], [3], [4] make it possible to hold the learner fixed and study the representation in isolation. The present project sits in this second family: the novelty is the hand-designed visual feature, while the reinforcement-learning algorithms are unmodified evaluators.

III. ENVIRONMENT AND PROBLEM DEFINITION

CarRacing-v3 [1] places an agent on a procedurally generated, randomly laid-out track. At each step the environment renders a top-down 96×96 RGB frame of the area around the car. The action space is continuous and three-dimensional:

$$a = [a_{\text{steer}}, a_{\text{gas}}, a_{\text{brake}}], \quad (1)$$

with $a_{\text{steer}} \in [-1, 1]$, $a_{\text{gas}} \in [0, 1]$, and $a_{\text{brake}} \in [0, 1]$. The reward is based on progress along the track and is reduced when the car visits tiles out of order or runs out of time. Tracks are randomized across episodes, so a successful policy must generalize rather than memorize a single layout.

Two properties make this a hard visual-control problem. First, the useful signal (the road and its local geometry) occupies only part of the frame, and it must be extracted from color and shape rather than given explicitly. Second, a single episode can look successful for many frames and then fail suddenly after a sharp turn, an off-road drift, or an unstable recovery. A single high-scoring video is therefore not sufficient evidence; the project evaluates saved artifacts on multiple fixed and randomized seed sets and reports mean, variance, and individual seeds.

The image-processing objective follows directly: convert each rendered frame into a small, inspectable description of the local road that an MLP policy can act on quickly and that generalizes across randomized tracks.

IV. IMAGE PROCESSING PIPELINE

The perception wrapper converts a raw RGB frame into a road mask and then into a set of ray features. Figure 1 summarizes the six stages.

A. Cropping and color conversion

The lower driving-relevant portion of the frame is retained and converted from RGB to HSV [7]. Working in HSV separates chromatic information (road versus grass) from intensity, which makes the road-region threshold more stable to overall brightness changes than an RGB threshold would be.

B. HSV road-mask extraction

The road is isolated by removing the grass. In the HSV representation the green grass occupies a well-defined band, so grass pixels are selected with a fixed lower and upper bound and the road mask is taken as the bitwise complement:

$$\text{grass}(p) = \mathbb{K}[l \preceq \text{HSV}(p) \preceq u], \quad \text{road} = \neg \text{grass}, \quad (2)$$

with $l = [28, 35, 35]$ and $u = [92, 255, 255]$ in OpenCV units ($H \in [0, 179]$, $S, V \in [0, 255]$) [7]. The hue window 28–92

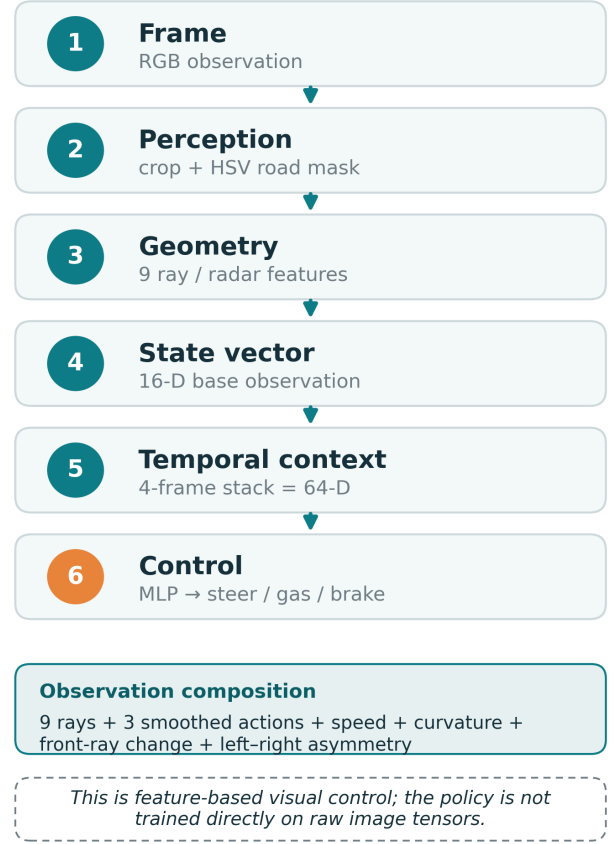


Fig. 1. Feature-based visual-control pipeline. A rendered RGB frame passes through six stages: frame (RGB observation), perception (crop and HSV road mask), geometry (nine ray/radar features), the 16-dimensional base state vector, temporal context (a four-frame stack, 64-dimensional), and control (an MLP emitting steering, gas, and brake). The base observation comprises nine rays, three smoothed actions, speed, curvature, a front-ray change term, and a left-right asymmetry term. The policy is not trained directly on raw image tensors.

covers the green range, while the saturation and value floors of 35 reject very dark or washed-out pixels that would otherwise be misclassified. Selecting the grass and complementing it, rather than thresholding the road directly, is more stable because the grass color is far more uniform than the combined road, curb, and car pixels.

The raw complement contains speckle and pin-holes, so it is cleaned with a 3×3 morphological opening—an erosion followed by a dilation with the same structuring element B :

$$\text{road} \leftarrow (\text{road} \ominus B) \oplus B. \quad (3)$$

The erosion removes isolated bright specks and thin bridges, and the dilation restores the boundary of the surviving regions, so the dominant road blob keeps its area while noise is discarded. Finally, a connected-component labeling pass can restrict the mask to the component containing the car's position, suppressing detached road-colored patches elsewhere in the frame.

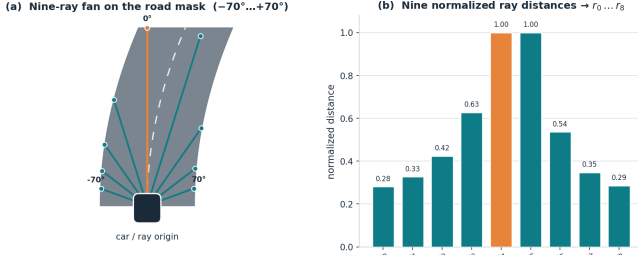


Fig. 2. Ray-feature extraction. (a) Nine rays are cast across the calibrated -70° to $+70^\circ$ fan and stop at the road-mask boundary, with the centre ray (0°) highlighted. (b) The travelled distances, normalized to $[0, 1]$, become the nine ray components $r_0 \dots r_8$ of the observation. The asymmetric profile shown here encodes a road bending to the right.

C. Ray casting

Local road geometry is summarized by nine distance rays cast from a calibrated origin near the car outward at fixed relative angles spanning a 140° forward fan:

$$\Theta = [-70, -52.5, -35, -17.5, 0, 17.5, 35, 52.5, 70] \text{ deg}, \quad (4)$$

where 0° points straight ahead and the sign denotes left ($-$) or right ($+$). Each ray is traced by stepping a point $\mathbf{p}(t) = \mathbf{p}_0 + t(\sin \theta, \cos \theta)$ outward in small fixed increments until it leaves the road mask or reaches a maximum length $L_{\max}$; the ray's value is the travelled distance normalized to $[0, 1]$,

$$r_i = \min(t_i^{\text{hit}}, L_{\max}) / L_{\max}. \quad (5)$$

The nine returns form a one-dimensional profile of how open the road is ahead and to each side (Fig. 2): a centred peak indicates a clear road ahead, while an asymmetric profile signals an upcoming bend. To keep the fan informative at speed, both the angular spread and $L_{\max}$ are modulated by the previous speed—the fan is gently narrowed (“squeezed”) and lengthened when the car is fast, giving a tighter, longer look-ahead, and widened when it is slow.

D. From mask and rays to a compact observation

The ray profile is the backbone of the observation, but it is augmented with control- and dynamics-oriented terms so the policy has enough state to act smoothly and recover. The curvature proxy is derived from the asymmetry between the left and right ray groups and the central ray depth; the front-ray change term measures the difference between the current central ray and its previous value, giving a local derivative of forward openness; and the left-right asymmetry term summarizes the bias of the road to one side. These are described in detail in Section V.

V. FEATURE REPRESENTATION DESIGN

A. Base observation vector

The base observation is a 16-dimensional vector:

$$\mathbf{o}_t = [r_0, \dots, r_8, \tilde{a}_{t-1}, v_t, \kappa_t, \Delta f_t, \alpha_t], \quad (6)$$

where $r_0, \dots, r_8$ are the nine normalized ray distances; $\tilde{a}_{t-1} \in \mathbb{R}^3$ are three smoothed previous action terms; v_t is the speed;

TABLE I
COMPOSITION OF THE 16-DIMENSIONAL BASE OBSERVATION.

Index	Component	Source
0–8	9 ray distances $r_0 \dots r_8$	ray casting on road mask
9–11	3 smoothed previous actions $\tilde{a}_{t-1}$	action history
12	speed v_t	environment state
13	curvature proxy κ_t	ray-group asymmetry
14	front-ray change Δf_t	central ray derivative
15	left-right asymmetry α_t	ray-group means

κ_t is the curvature proxy; Δf_t is the front-ray change term; and α_t is the left-right asymmetry term. The composition is summarized in Table I.

The curvature proxy is active only when the forward road is judged to be present (the mean of the central forward rays exceeds a small threshold), which prevents the curvature estimate from responding to off-road noise.

Each augmenting term carries a specific control role. The three smoothed previous actions $\tilde{a}_{t-1}$ give the policy access to its own recent commands, which discourages jittery steering and lets a maneuver continue smoothly. The speed v_t disambiguates otherwise identical ray profiles that demand different throttle or braking. The curvature proxy κ_t and the left-right asymmetry α_t pre-compute the direction and magnitude of the turn from the ray groups, so the network need not re-derive them. The front-ray change Δf_t is a one-step derivative of forward openness that gives early warning of an approaching edge before the absolute distance becomes small.

B. Temporal stacking

A single frame’s observation is instantaneous and cannot express motion. Following standard practice, $N = 4$ consecutive observations are stacked to form the policy input:

$$\mathbf{s}_t = [\mathbf{o}_{t-3}, \mathbf{o}_{t-2}, \mathbf{o}_{t-1}, \mathbf{o}_t] \in \mathbb{R}^{64}. \quad (7)$$

The stacked 64-dimensional vector carries short-horizon dynamics (velocity of the rays, change in curvature, action momentum) while remaining far smaller and more inspectable than a stack of raw image tensors.

C. Shared representation across algorithms

A central design decision is that the perception layer is *held constant*. The 16-dimensional base observation and the 64-dimensional stacked observation are identical for both downstream evaluators. PPO and SAC therefore see exactly the same visual feature stream; the only difference between the two experiments is the policy optimization algorithm. This isolates the image-processing contribution as the common variable and makes any behavioral difference attributable to the learner rather than to the representation.

D. Reward and action shaping

The environment’s raw reward (track progress) is the headline metric and is the quantity reported in all results. During training only, a shaped reward combines the raw reward with

auxiliary terms (for example on-road incentives and anti-stagnation terms) to stabilize early learning. Shaped reward is a training aid; all reported numbers are raw rewards. The action is also smoothed so that the policy emits continuous, non-jittery commands.

VI. REINFORCEMENT LEARNING EVALUATION SETUP

Both learners are off-the-shelf Stable-Baselines3 algorithms with MLP policies [2], [3], [4]. No algorithm is modified. The wrapper-defined compact observation (Section V) is wrapped in a vectorized environment with observation normalization (VecNormalize). Checkpoint-based evaluation callbacks record the best mean reward during training and save the matching model and normalization statistics.

A. Checkpoint and reload logic

Evaluation is artifact-driven. At evaluation time, the saved model parameters are restored from a saved checkpoint and the saved VecNormalize statistics are reloaded so that observation normalization at evaluation matches training. The model is reloaded without reattaching the training environment, which keeps the evaluation seed schedule fixed and prevents it from being overwritten during reload. For the SAC fast-result branch, the saved 400,000-step checkpoint is used as the fast-result best checkpoint.

B. Evidence hierarchy

Headline metrics are parsed from the training logs and the validated best checkpoints. A secondary tier of evidence is computed from saved notebook outputs on a fixed multi-seed evaluation set and a randomized multi-seed evaluation set, with a bootstrap confidence interval. These secondary values are reported separately and are never confused with the log-validated headline metrics.

VII. PPO BASELINE EXPERIMENT

The PPO baseline [5] is a completed 500,000-step run. Throughput was approximately 40 frames per second (elapsed roughly 12,417 seconds). The final evaluation at 500,000 steps reaches a raw reward of 938.87 ± 7.86 , and the best parsed checkpoint is 939.53 ± 4.09 at 480,000 steps. The full per-checkpoint curve is reported in the Appendix (Table V).

The curve shows the expected noisy climb through the middle of training, followed by a clear stabilization into the low-930s to low-940s band over the final checkpoints. The last observed training step was 501,760, confirming that the run reached the full budget.

VIII. SAC FAST-RESULT BRANCH

The SAC fast-result branch [6] is a *partial* run. Its validated result is the 400,000-step best checkpoint, which reaches a raw reward of 938.51 ± 4.88 . The last observed training step was 418,697 (roughly 82% of 500,000), and there is no evaluation checkpoint after 400,000 in the provided log; steps beyond 400,000 are therefore not treated as a validated best result. Throughput was approximately 36 frames per second (elapsed

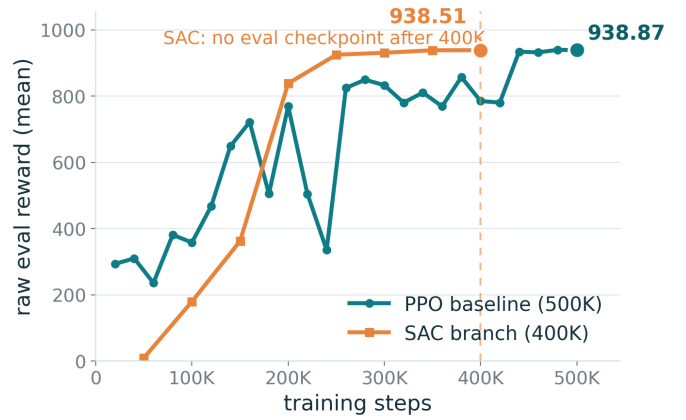


Fig. 3. Training curves of the PPO baseline and the SAC fast-result branch, plotted from the checkpoint tables. SAC stops at the validated 400,000-step checkpoint.

TABLE II
HEADLINE COMPARISON. PPO IS A COMPLETED 500K BASELINE; SAC IS A PARTIAL 400K BEST-CHECKPOINT FAST-RESULT BRANCH. THE COMPARISON IS NOT COMPUTE-EQUIVALENT.

Branch	Status	Validated eval
PPO baseline	Completed baseline	500K 938.87 ± 7.86 @ 500K; best 939.53 ± 4.09 @ 480K
SAC branch	Partial 400K best checkpoint	938.51 ± 4.88 @ 400K

roughly 11,400 seconds). Early stopping reflects Colab run-time limits and the fast-result intent of the branch. The full per-checkpoint curve is reported in the Appendix (Table VI).

IX. RESULTS AND ANALYSIS

A. Headline comparison

Table II summarizes the headline comparison. Both learners reach the low-930s to low-940s band on the shared compact representation, which is the main qualitative finding: the same image-processing-derived feature stream supports strong driving behavior under two different off-the-shelf policy optimizers.

B. Checkpoint and compute budget

Figure 3 shows the two training curves. PPO climbs noisily and then stabilizes; SAC converges faster in raw-reward terms within its partial budget. Because the two runs differ in validated budget (500K versus 400K), throughput, and total elapsed time, the curves are descriptive and not a controlled ablation. The compute budget is disclosed in Table III.

C. Secondary generalization evidence

Beyond the log-validated headline metrics, saved notebook outputs provide secondary generalization evidence on a fixed multi-seed evaluation set, a randomized multi-seed evaluation set, and a bootstrap confidence interval. These values are summarized in Figure 4 and Table IV and are explicitly

TABLE III
COMPUTE AND BUDGET DISCLOSURE.

Branch	Validated budget	Throughput / elapsed
PPO baseline	500,000 (completed)	~40 fps / 12,417 s
SAC branch	400,000 (partial)	~36 fps / 11,400 s

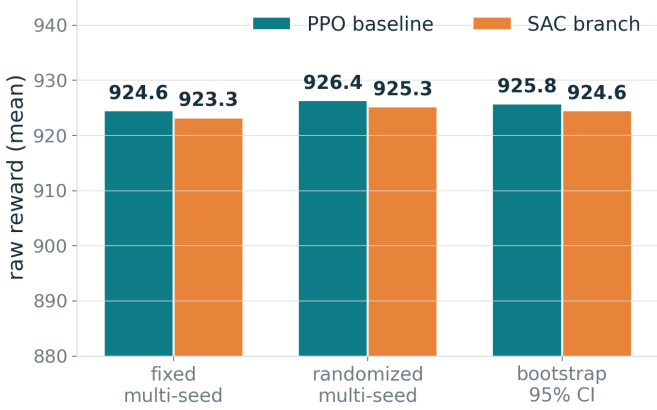


Fig. 4. Secondary generalization evidence from saved notebook outputs (fixed multi-seed set, randomized multi-seed set, and bootstrap confidence interval). This is notebook-output evidence, distinct from the log-validated headline metrics.

secondary: they are computed from saved notebook outputs and must not be read as log-validated headline numbers.

Both branches land in the mid-920s on the held-out seed sets—a few points below the best single checkpoints in Table II, as expected when averaging over harder randomized layouts—and the bootstrap confidence intervals are narrow, which indicates the mean behavior is stable even though individual seeds vary.

X. DISCUSSION AND LIMITATIONS

The central result is that an image-processing-derived compact representation, rather than raw pixels, can support strong driving behavior with off-the-shelf learners and an MLP policy. Holding the representation constant across PPO and SAC isolates the perception contribution: both learners reach the same reward band on the same feature stream, which is consistent with the hypothesis that the representation—not the learner—is doing the heavy lifting in this setting.

Several limitations bound these conclusions. First, the SAC result is a partial fast-result branch validated only at the 400,000-step checkpoint; it did not complete 500,000 steps, so the PPO/SAC comparison is not compute-equivalent and SAC is not claimed to beat PPO. Second, no raw image-input CNN policy was trained in this submission, so the compact representation is not benchmarked against a pixel baseline here. Third, HSV-based road-mask extraction depends on color appearance and can be sensitive to rendering or track-art changes; the method is not claimed to be uniformly robust. Fourth, the project does not claim that CarRacing-v3 is solved.

Failure modes. The aggregate rewards hide a heavy-tailed failure distribution. On most seeds the policies stay in the

TABLE IV
SECONDARY GENERALIZATION EVIDENCE (SAVED NOTEBOOK OUTPUTS), ON A FIXED MULTI-SEED SET, A RANDOMIZED MULTI-SEED SET, AND A BOOTSTRAP 95% CONFIDENCE INTERVAL. REPORTED SEPARATELY FROM THE LOG-VALIDATED HEADLINE METRICS.

Branch	Split	Mean $\pm$ std
PPO baseline	fixed seeds	924.6 $\pm$ 6.8
PPO baseline	random seeds	926.4 $\pm$ 7.3
PPO baseline	bootstrap CI	925.8 [922.2, 929.0]
SAC branch	fixed seeds	923.3 $\pm$ 7.7
SAC branch	random seeds	925.3 $\pm$ 8.2
SAC branch	bootstrap CI	924.6 [920.7, 928.2]

low-930s to low-940s band, but a small number of seeds fail badly: two representative failure seeds, 599310825 and 440213415, score only about 55–56 raw reward. These cases are typically triggered by a sharp bend that the forward fan does not resolve in time, or by an unstable recovery after a brief grass excursion, during which the connected-component stage momentarily loses the road region. This long tail is the main reason the method is described as promising rather than uniformly robust, and it is why results are reported as multi-seed statistics rather than a single best episode.

XI. CONCLUSION

This project presented an image-processing-centric pipeline for CarRacing-v3 in which rendered RGB frames are converted into a compact, inspectable road-mask ray/radar feature representation. The representation—a 16-dimensional base observation temporally stacked into a 64-dimensional vector—is shared by a completed PPO baseline (938.87 ± 7.86 at 500,000 steps; best parsed 939.53 ± 4.09) and a SAC fast-result branch (938.51 ± 4.88 at the 400,000 best checkpoint). The reinforcement-learning algorithms are unmodified Stable-Baselines3 evaluators; the contribution is the visual representation. The method is sample-efficient and promising, but the SAC result is partial, the PPO/SAC comparison is not compute-equivalent, and CarRacing-v3 is not claimed to be solved. Future work includes matched-budget multi-seed runs, a raw-pixel baseline, and systematic ablation of the perception stages.

APPENDIX

Tables V and VI list the complete per-checkpoint evaluation curves for the two branches, parsed from the training logs and summarized in the main text.

The notebooks are kept in safe report/evaluation mode (SAFE_MODE, REPORT_ONLY, EVAL_ONLY enabled and training disabled by default). Headline metrics are parsed from the training logs and validated best checkpoints. The full checkpoint tables (Tables V and VI) are the primary numerical evidence; the secondary generalization evidence is computed from saved notebook outputs.

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TABLE V
PPO BASELINE: RAW REWARD (MEAN $\pm$ STD) AT EVALUATION
CHECKPOINTS, PARSED FROM THE TRAINING LOG.

Step	Reward	Step	Reward
20,000	293.54 $\pm$ 146.56	280,000	849.36 $\pm$ 173.34
40,000	310.22 $\pm$ 89.79	300,000	832.78 $\pm$ 204.18
60,000	236.05 $\pm$ 79.57	320,000	780.22 $\pm$ 211.17
80,000	381.07 $\pm$ 69.73	340,000	810.58 $\pm$ 233.17
100,000	358.19 $\pm$ 132.66	360,000	768.91 $\pm$ 211.34
120,000	467.53 $\pm$ 211.29	380,000	857.28 $\pm$ 104.10
140,000	648.81 $\pm$ 243.53	400,000	784.98 $\pm$ 178.68
160,000	721.70 $\pm$ 135.85	420,000	780.46 $\pm$ 317.26
180,000	505.19 $\pm$ 296.11	440,000	934.05 $\pm$ 6.67
200,000	769.12 $\pm$ 223.36	460,000	931.94 $\pm$ 2.33
220,000	505.04 $\pm$ 355.49	480,000	939.53 $\pm$ 4.09
240,000	335.40 $\pm$ 109.84	500,000	938.87 $\pm$ 7.86
260,000	824.59 $\pm$ 182.66		

TABLE VI
SAC BRANCH: RAW REWARD (MEAN $\pm$ STD) AT EVALUATION
CHECKPOINTS, PARSED FROM THE TRAINING LOG. NO CHECKPOINT
EXISTS AFTER 400,000.

Step	Reward (mean $\pm$ std)
50,000	9.67 $\pm$ 8.98
100,000	179.94 $\pm$ 28.69
150,000	362.34 $\pm$ 53.22
200,000	838.10 $\pm$ 151.24
250,000	924.99 $\pm$ 4.55
300,000	930.68 $\pm$ 6.09
350,000	938.33 $\pm$ 2.42
400,000	938.51 $\pm$ 4.88

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